

Weekly Report 5

Project title: Scot Forge

Date: 03/08/2026

Team members:

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Progress Report:

- Presented different forecasting models along with their evaluation metrics during the call with Mr. Fass. (Kumar, Pratyaksh, Vandit)
- Discussed the reasoning behind selecting an appropriate primary evaluation metric for model comparison. (Vandit, Kumar)
- Explained the Walk-Forward validation approach used for evaluating the models. (Pratyaksh)
- Reviewed the natural gas tariff documentation, focusing on transportation and storage provisions.
- Discussed daily storage activity parameters, including allowed injection and withdrawal percentages based on total storage capacity.
- Reviewed monthly storage inventory balance requirements, which define the allowable storage range at the end of each month.
- Learned about the daily and monthly cash-out penalty mechanisms, including the tiered penalty structure applied when operational limits are exceeded.

Action Items:

- Calculate daily injections and withdrawal limits based on the total storage capacity.
- Understand the penalty and cash-out system in detail to determine operational hard caps.
- Evaluate how the existing models perform under the storage and penalty constraints.
- Optimize the models incorporating these operational constraints.